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## **Novel Application of Distributed Temperature Sensing and Downhole Sensors Provides Unprecedented Confidence to Operate Conventional Sliding Sleeves in Horizontal Wells: Case Study from the United Arab Emirates**

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### **Abstract**

Sliding sleeve doors (SSDs) are widely used in well completions as flow management devices that allow selective communication between the production string and annulus. Although the industry has progressed with "smart" completions that allow SSD operations from surface, in many cases, SSDs continue to be operated mechanically using slickline (SL).

For deviated wells or difficult environments (i.e., reservoirs with heavy oil production) where SL operations are unsuitable, coiled tubing (CT) is required to help ensure safe access to the target depth. The use of conventional CT units to operate SSDs is widely adopted globally. However, service quality issues are caused by several uncertainties, such as inaccurate depth correlation based on CT friction wheels, effective downhole force (calculated as a function of the surface weight), and the requirement to run a production logging test after CT operation to verify the final SSD position.

A case history is discussed wherein fiber-optic-equipped CT was deployed with a modular sensing bottomhole assembly (MSBHA) to operate SSDs with real-time downhole confirmation (tension and compression) followed by interval flow confirmation through distributed temperature sensing (DTS) in a single run. This approach reduced operational time and dramatically increased confidence in the results.

The operation incorporated a combination of multiple technologies:

- Casing collar locator (CCL) for accurate depth correlation
- Tension and compression data to confirm SSD activation and eliminate uncertainties related to string weight and friction losses
- Internal and external pressure and temperature measurements to identify immediate reservoir responses

Data from the bottomhole assembly (BHA) was then combined with the DTS analysis to provide a complete injection profile of the well before and after SSD manipulation. This verified fluid intake at each interval to confirm the operational status of each SSD.

Downhole sensors provided superior accuracy in depth measurement and forces applied on the SSDs, while the DTS identified a distinct pattern of high- and low-injectivity zones. Flow comparisons before and after the SSD operation confirmed the device position, increased operational efficiency, and removed the requirement for additional confirmation runs.

This case history represents a step-change in SSD operations in complex environments. The solution provided confidence and optimization in a single run, saved operational time, and ensured execution of the desired production arrangement as per the intervention objectives.

## Introduction

SSDs are downhole valves, also known as inflow control devices (ICDs), that facilitate or shut off communication between the tubing and annulus in a wellbore. In horizontal or highly deviated wells, confirmation of SSD status (opened or closed) presents a challenge because conventional wireline SL intervention is impractical. CT-conveyed tools provide a solution that physically reaches these zones (Daniels and Williams 2001). Real-time, fiber-optic coiled tubing (RTFOCT) provides a method to increase confidence in the SSD operation. The RTFOCT system integrates a 4-mm fiber-optic cable within the CT string that contains six fibers—one for distributed acoustic sensing (DAS) and five for DTS, telemetry, and redundancy. This configuration allows continuous data transmission from downhole sensors and optional cameras. Surface systems, such as a slip ring and data acquisition units, maintain uninterrupted power and real-time data flow.

When equipped with a MSBHA that includes tension, compression, and CCL sensors, the RTFOCT system delivers real-time downhole mechanical data during SSD shifting operations. In tandem, a DTS fiber-optic system delivers continuous temperature profiles along the wellbore, which provides indirect confirmation of flow changes when SSDs open or close (Al Matar et al. 2020).

During the actual shifting process, the interplay of the sensors produces distinct signals that correlate with SSD movement. These signal signatures form the basis for the confirmation of the SSD position (open or closed) purely from downhole measurements:

- Initial latch signature
- Force rampup
- Break and drop
- End-of-travel and key release
- Verification pass

These mechanical sensor signatures serve as a direct confirmation of actuation, and the mechanical data assures the SSD has physically moved as intended.

## Downhole Sensor Suite for SSD Actuation

Modern CTMSBHAs can be outfitted with a suite of downhole sensors that deliver crucial feedback during SSD shifting operations. Key components include (Lerose et al. 2020):

- CT Connector: Secures the connection between the CT and BHA.
- Cablehead Assembly: Provides fiber connectivity and permits data transmission between surface systems and downhole sensors.

- Sensor Module: A compact, 2.25-in.OD unit equipped with:
  - CCL
  - Internal and external pressure sensors
  - Internal and external temperature sensors
  - Tension, compression, and torque sensors
- Motorhead Assembly: Features a rupture disk, double-flapper check valves, and a hydraulic disconnect for safety and contingency.

This modular BHA is compatible with mechanical shifting tools and is rated for well operations with temperature up to 350°F and pressure up to 15,000 psi. These sensors function synergistically to guide and confirm SSD manipulation as described in subsequent sections.

## Tension and Compression Sensors

The tension and compression sensors measure the axial force applied to downhole tools in both tension (pull) and compression (push) modes. The primary function is to monitor set-down weight, or weight on bit (WOB), which, in this application, represents the force used to shift the SSD. During an SSD shifting operation, the CT tool latches onto the shifting profile of the sleeve. Once latched, the applied force registers on the sensor.

### Tension Force (Overpull)

For an SSD with a shift-up-to-open design, the CT operator lifts, which creates tension in the string. The downhole tension sensor records the increased pull as the sleeve's lock mechanism resists movement. The force typically ramps up to a peak once the resistance threshold is reached.

When the sleeve breaks free and begins to slide open, the tension measurement drops suddenly. This sudden decrease is a characteristic signature that indicates sleeve movement and the release of resistance. In practice, an overpull spike followed by a sharp decrease in load confirms the SSD has shifted to the open position.

Once open, the tool's shifting keys often disengage from the sleeve profile. This disengagement appears as a distinct reduction in force or a noticeable "free-up" of the string, which provides another positive indication of a successful shift.

### Compression Force (Set-Down Weight)

Conversely, if the SSD requires a shift-down-to-close or shift-down-to-open action, the operator slacks off weight and places the BHA in compression. The compression sensor then measures the set-down weight applied against the sleeve. A similar signature is expected—the compressive force increases as the tool pushes on the sleeve, which peaks at the point of mechanism release, followed by a sudden drop when the sleeve snaps to the closed or open position. This drop signifies the internal sleeve has fully shifted and resistance has been relieved. Accurate compression measurement is important in deviated wells because friction can mask the true weight-on-tool at the surface. A downhole sensor helps ensure the correct force is delivered to actuate the SSD.

Tension and compression readings are critical for SSD mechanism interaction. The SSD is typically held in place by lock catches or shear pins that require a specific force to overcome. The downhole load sensor directly records the force profile as the shift tool engages the SSD. A "force vs. displacement" signature is obtained and exhibits a steady increase in force (elastic load) until the sleeve moves, followed by an immediate drop once movement occurs. This real-time feedback helps the operator confirm that the required force was applied and the sleeve responded as intended.

## CCL

The CCL functions as a depth correlation tool that emits a magnetic field and detects changes in metal volume within the wellbore. It identifies casing or tubing collars by the minor increase in metal thickness at those points. When the CCL passes a collar, the disturbance in the magnetic flux induces a voltage spike in the tool's coil sensor. These spikes, spaced at known intervals, allow precise depth determination ([Vera et al. 2019](#)).

In the context of an SSD operation, the CCL pinpoints the location of the sliding side door within the completion. Using the measured collar count to align with the well's completion tally, the operator can position the shifting tool exactly at the SSD's depth before an open or close action begins.

## Downhole Pressure (External/Internal)

Downhole pressure (external and internal), in addition to tension, compression, and CCL correlation, plays a critical role in the improvement of SSD or ICD shifting operational accuracy. Internal pressure measurements allow operators to monitor the annular or tubing pressure inside the CT string. This is essential for the evaluation of hydraulic force application, tool activation, and verification of BHA integrity during pump operations ([Corrales et al. 2024](#)). On the contrary, external pressure sensors provide direct feedback on the formation or wellbore pressure outside the tubing, which allows real-time assessment of differential pressure in sleeve ports or ICD nozzles.

This differential pressure ( $\Delta P$ ) is crucial to determine if a device has successfully opened or closed, as the sensor is positioned directly above the shifting tool. Changes in external pressure behavior, such as pressure stabilization or anomaly detection, can indicate flow initiation or restriction. Furthermore, a comparison of internal and external pressure responses during injection or flowback provides wellbore dynamics, confirms fluid placement, and identifies potential sealing issues or crossflow between compartments. Together with mechanical force signatures and DTS interpretation, these pressure measurements form a resilient diagnostic triad that helps ensure comprehensive SSD operation validation in real time.

## DTS-Based Thermal Anomaly as Indirect SSD Validation

While surface mechanical sensors and downhole tension/compression data confirm an SSD's sleeve shift (together with a confidence added on the depth correlation conducted with the CCL), an additional confirmation layer can be obtained with an observation of changes in the well's thermal profile through DTS. DTS involves a fiber-optic line run along the well that measures temperature continuously. When an SSD is opened, it alters the flow distribution. This flow change manifests as a local temperature disturbance. Therefore, DTS acts as a real-time flow assessment tool that provides validation of SSD position by the presence or absence of flow-induced thermal signatures ([Al Sebea et al. 2023](#)).

The DTS survey normally has the following three distinct phases.

### Baseline Temperature Trace

The well was shut in to allow thermal stabilization. This phase was essential to establish the geothermal baseline, which would serve as a reference to identify temperature anomalies during and after fluid injection.

### Cool-Down Phase

Injection water is bullheaded through the CT/completion annulus at a controlled rate. The objective is to induce a thermal contrast in the formation, which permits the identification of active injection zones through DTS temperature analysis. Temperature changes were recorded to identify zones of active fluid ingress. The cooling effect was expected to highlight areas of fluid ingress and reservoir interaction.

## Warm-Back Phase

After injection cessation, the well was returned to the shut-in mode. The DTS system continued to monitor temperature recovery over several hours. This phase provided insights into the thermal behavior of the formation and validated fluid movement in the ICDs. The rate and extent of warm back indicated formation conductivity and fluid retention characteristics.

## Operation Execution

The operation began with a runinhole (RIH) to total depth (15,543 ft) using a BHA equipped with a CT connector, anchor section, MSBHA, motorhead assembly, and rotating nozzle. Recorded RIH weights were within the expected range. However, a significant increase in pullout of hole (POOH) weight to 42,000 lbf indicated possible scale buildup or debris. This observation prompted intermediate tool checks and partial POOH runs to confirm tool freedom. After mechanical integrity confirmation, the BHA was positioned below the first swell packer to initiate a DTS baseline (Fig. 1).

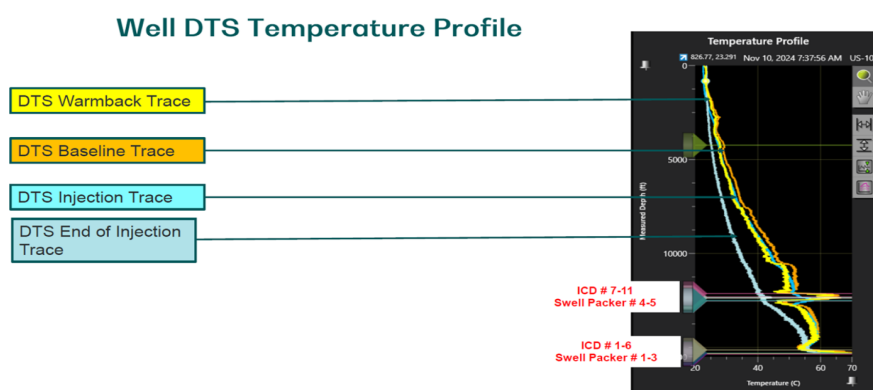


Figure 1—Well DTS profile before shifting operation.

With the well shut in, a thermal stabilization period established a reference temperature profile. A controlled water injection at 2.9 bbl/min followed, which injected approximately 450 bbl to induce a cooling phase. The subsequent warm-back period under shut-in conditions provided thermal relaxation data to evaluate formation heat retention and flow paths. The DTS profile indicated a significant temperature drop during injection and steady thermal recovery after shut in (Fig. 2).

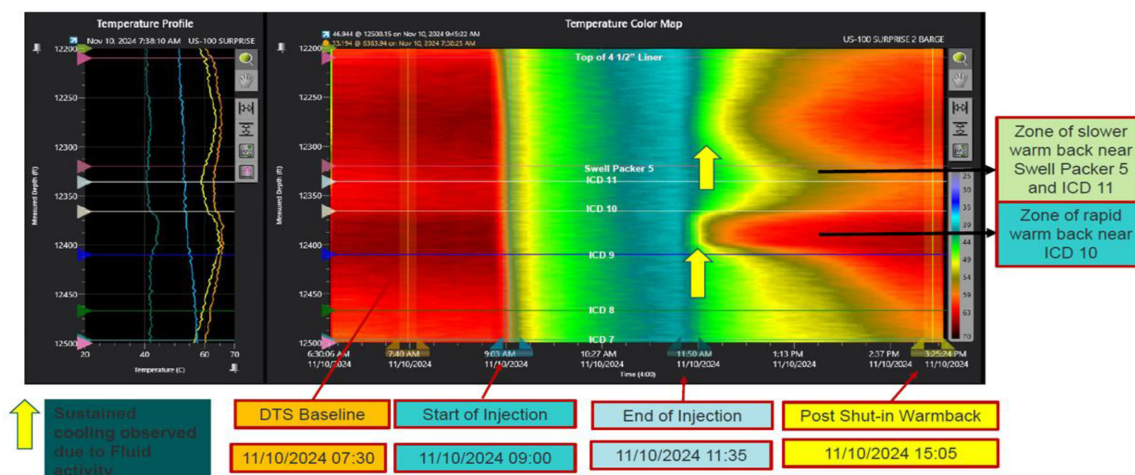


Figure 2—DTS diagnostics stages before shifting operation.

A targeted cleanout between 15,380 and 15,475 ft was executed using a rotating jetting tool. The operation applied 12 bbl of organic dissolver and 30 bbl of metal-to-metal friction reducer to remove scale and improve flow efficiency. A four-hour soak period allowed adequate chemical reaction time. This cleanout was critical to prepare the ICDs for a reliable shifting through the removal of mechanical blockages and a reduction of friction. The CT string was POOH to surface, and the rotating nozzle BHA was replaced with a shifting BHA.

The high-expansion shifting tool (HEST) was assembled and hydraulically calibrated in surface tests at three pump rates: 0.3 bbl/min at 1,000 psi, 0.6 bbl/min at 2,400 psi, and 0.8 bbl/min at 5,000 psi. The calibration established 0.6 bbl/min with a four-minute activation time as the optimal setting. These calibration steps helped ensure accurate hydraulic force delivery for SSD engagement during downhole operations. Fig. 3 shows the full BHA.

| Bottom Hole Assembly      |                                                                                                                                                                                                                           |                                                                                    |           |          |             |             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------|----------|-------------|-------------|
| Item                      | Tool Description                                                                                                                                                                                                          |                                                                                    | Tool O/D  | Tool I/D | Length (in) | Connection  |
| 1                         | CT Internal Dimple Connector                                                                                                                                                                                              |  | 2.125     |          | 5.00        | 1-1/2" AMMT |
| 2                         | ANCHORING SECTION                                                                                                                                                                                                         |                                                                                    | 2.250     |          | 173.04      | 1-1/2" AMMT |
|                           | SENSOR SECTION                                                                                                                                                                                                            |                                                                                    |           |          |             |             |
|                           | CPAT<br>Tension Compression Module<br>Battery Module<br>CCL Module<br>- Max Tension - 40k Lbs<br>- Max Compression - 10k Lbs<br>- Max Passable Ball - 9/16"                                                               |                                                                                    |           |          |             |             |
| 3                         | MHA<br><u>Hyd Disconnect - 0.5" Ball</u><br>Pressure - 3 Screws * 1409 psi =<br>4227 psi disconnect pressure<br><br><u>Circulation Sub - 0.4375" Ball</u><br>Pressure - 4 Screws * 733 psi =<br>2932 psi Release pressure |                                                                                    | 2.125     |          | 36.00       | 1-1/2" AMMT |
| 4                         | High Expansion Shifting Tool ( HEST)<br><br>Keys OD Expanded - 3.45"                                                                                                                                                      |                                                                                    | 2.200     |          | 53.00       | 1-1/2" AMMT |
| Total Length of B.H.A. :- |                                                                                                                                                                                                                           | 22.25 ft                                                                           | 267.04 in |          |             |             |
| Max OD :                  |                                                                                                                                                                                                                           | 2.25 in                                                                            |           |          |             |             |

Figure 3—BHA shifting tool and sensors.

The CT was RIH to 12,600 ft to initiate depth correlation using the CCL (Fig. 4). An offset of 28.37 ft was determined and applied to align the CT depth counter with the actual tool depth.



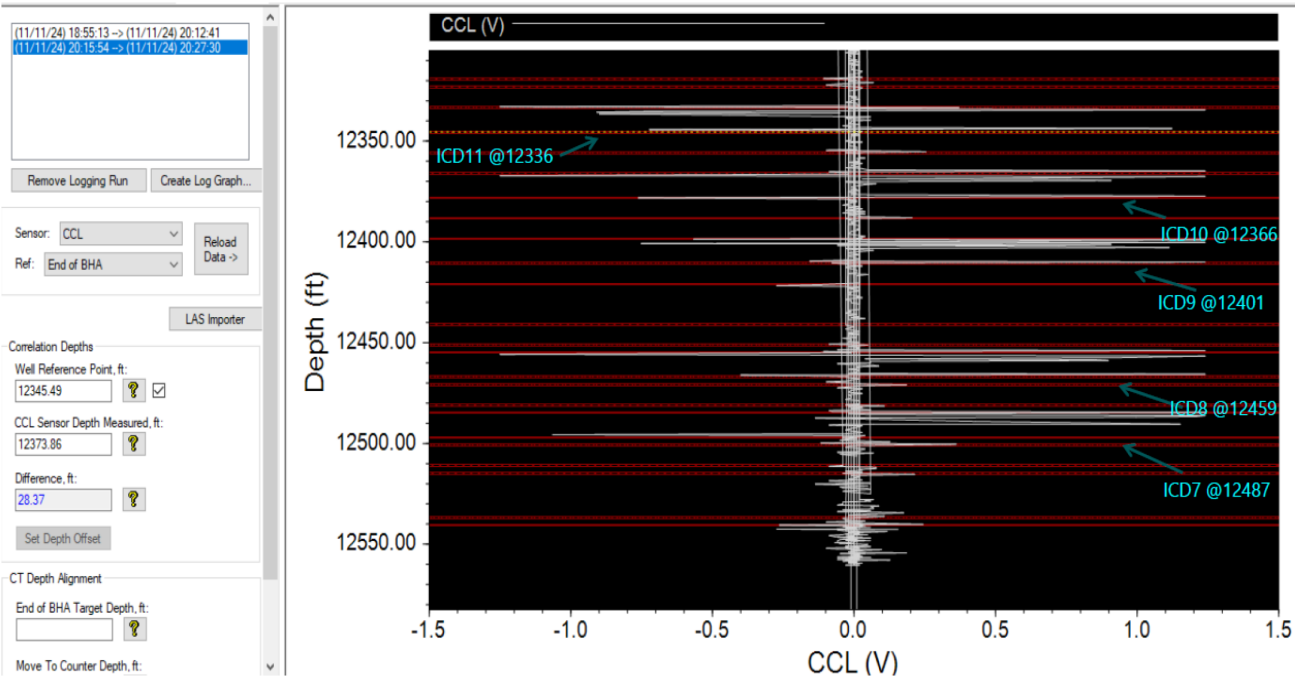


Figure 4—CCL depth correlation.

The shifting process began 15 ft below ICD 10 at 12,405 ft. A pump rate of 0.6 bbl/min activated the HEST, and real-time tension measurements confirmed mechanical engagement. ICD 10 produced a 4,000-lbm overpull, which indicated a positive sleeve shift. ICD 11 registered a 2,000-lbm signature.

Subsequent free-pass runs confirmed that the sleeves were in the closed position, verified by the absence of mechanical resistance. No shear pins failed during POOH, which confirmed successful nondestructive shifts. Fig. 5, Fig. 6, and Fig. 7 provide downhole sensor measurements.

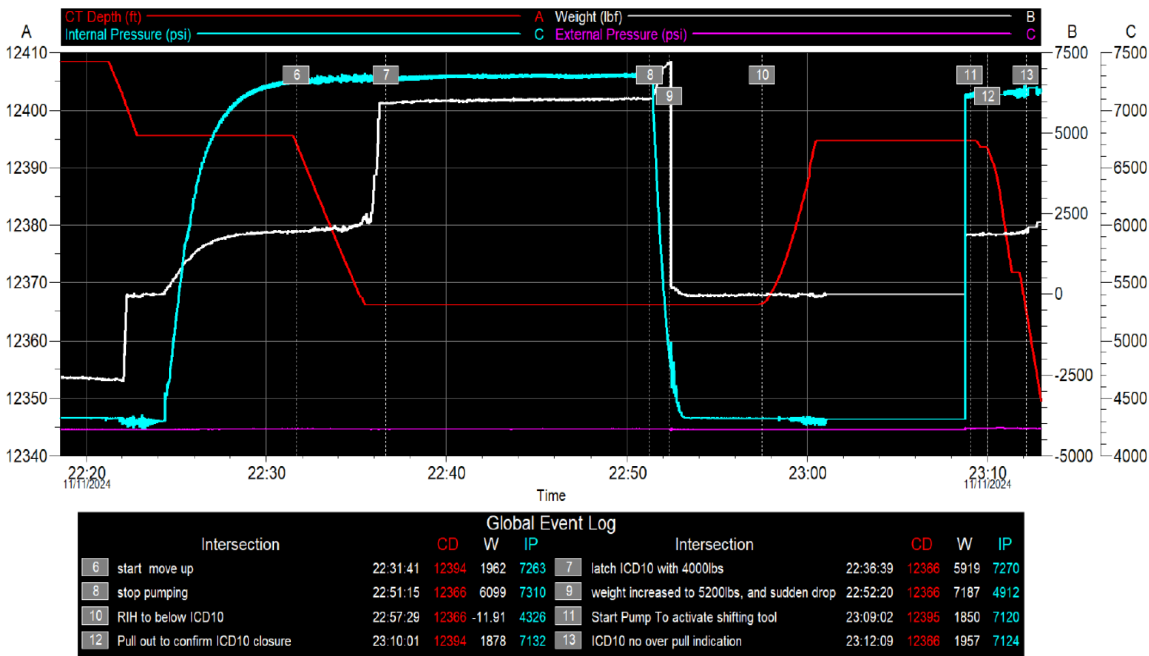


Figure 5—Downhole sensors reading: ICD 10 operation.

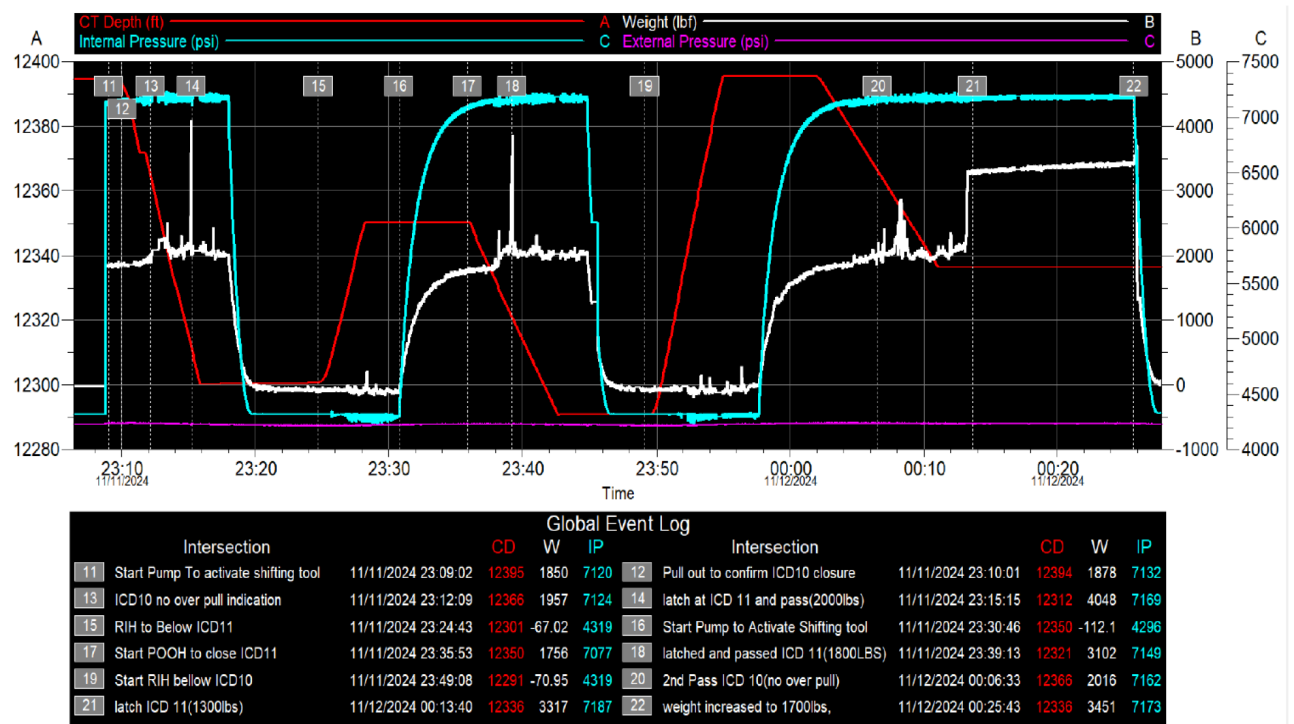


Figure 6—Downhole sensor measurement reading: ICD 10 cont. and ICD 11 operation.

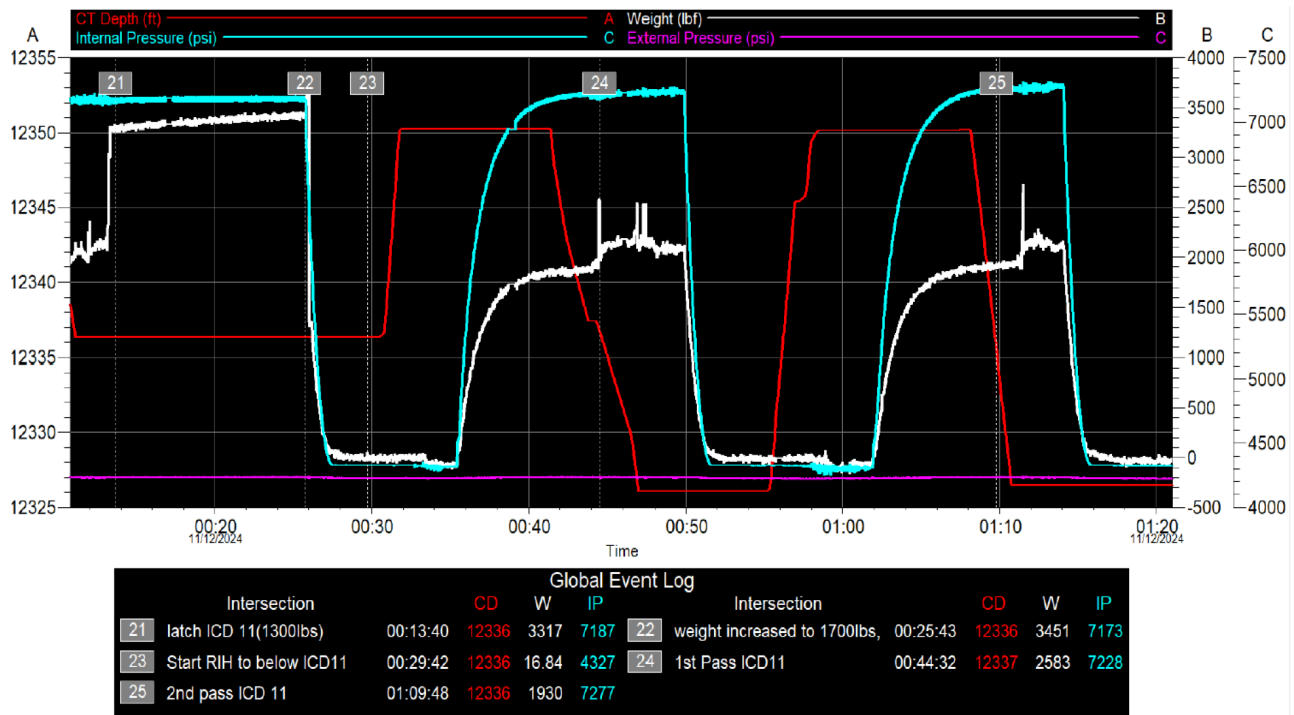


Figure 7—Downhole sensor measurements: ICD 11 operation.

A post-shifting DTS survey was executed with the CT positioned at 14,000 ft. As before, baseline traces, cooldown through injection, and warm-back phases were monitored. Thermal data were analyzed to detect flow behavior changes after ICD closures.

Rapid cooling was observed to 13,954 ft (fiber terminal depth), which confirmed active injection. However, fiber depth limits prevented the evaluation of zones below this point. A follow-up injection phase



assessed the effect of ICD closures. ICDs 10 and 11 were closed, while 7, 8, and 9 remained open. DTS confirmed cooling at the open ICDs and within the openhole horizontal drain section.

Subtle cooling near ICD 11 suggested possible crossflow or minor leakage, which indicates compartmental communication or incomplete isolation. In contrast, ICD 10 exhibited minimal thermal change, consistent with low injectivity before closure. Thermal recovery profiles returned to baseline after shutin, which confirmed transient behavior and validated completion performance (Fig. 8 and Fig. 9).

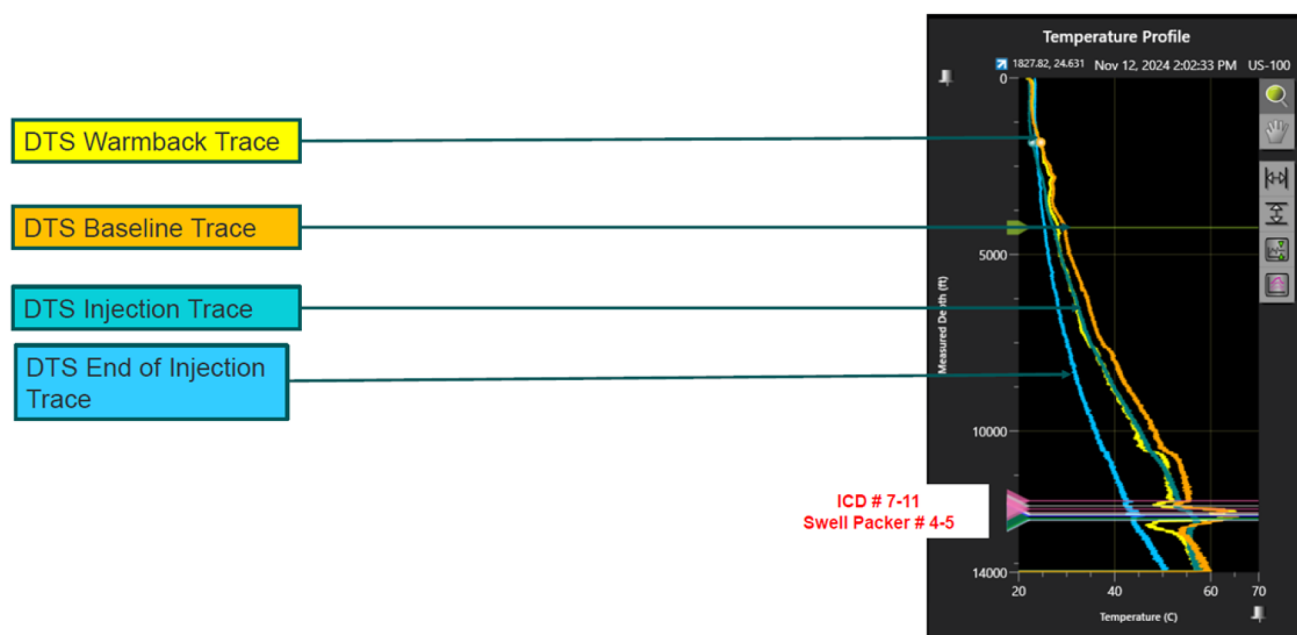


Figure 8—Well DTS profile after shifting operation.

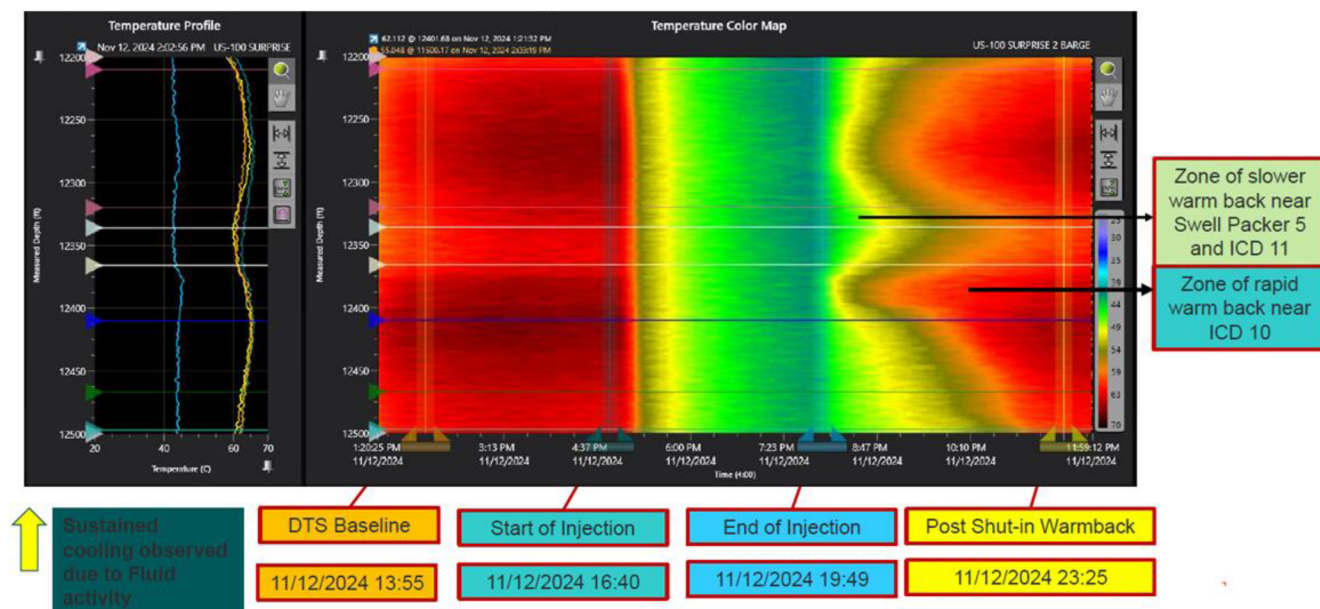


Figure 9—DTS diagnostics stages after shifting operation.

## Downhole Sensors and DST Synergy for SSD Operation Validation

Tension and compression sensor data conclusively demonstrated successful sleeve actuation based on characteristic overpull signatures. Depth correlation using the CCL helped ensure precise target alignment, and DTS provided thermal confirmation of flow redistribution.

This integrated approach allowed high-confidence ICD shifts in a horizontal wellbore without wireline access. The cleanout, hydraulic calibration, and real-time diagnostics contributed to the overall success of the intervention. Despite DTS depth limitations, the combined mechanical and thermal data supported the effectiveness of the shifting operation and provided valuable insight into reservoir response.

## Conclusion

This case exemplifies the synergy of mechanical force data and DTS diagnostics to confirm sleeve movement in CT operations. The combination of tension and compression signatures, depth correlation, and DTS thermal mapping provided a complete assessment of intervention results. In horizontal wells with restricted SL access and significant uncertainty using conventional CT units, this methodology provides a reliable and repeatable strategy for downhole device actuation and flow verification. The integration of mechanical sensor signals with DTS observations creates a comprehensive view of SSD actuation in complex conditions. By eliminating the need for additional intervention runs caused by uncertainties during execution, this approach effectively reduces operational time and associated costs.

The use of downhole sensors—tension, compression, pressure, and CCL—in conjunction with DTS during CT operations delivers an effective method to identify and confirm the proper opening or closing of SSDs in horizontal wells. The tension and compression sensors directly capture the mechanical signature of an SSD shifting event, the CCL helps ensure accurate positioning, downhole pressure measurements provide instantaneous readings of reservoir connection to the well, and the DTS fiber offers a continuous indicator of thermal anomalies as an indirect yet reliable indicator of flow through the SSD.

## Nomenclature

|        |                                     |
|--------|-------------------------------------|
| BHA    | bottom hole assembly                |
| CCL    | casing collar locator               |
| CT     | coiled tubing                       |
| DAS    | distributed acoustic sensing        |
| DTS    | distributed temperature sensing     |
| HEST   | high-expansion shifting tool        |
| ICD    | inflow control device               |
| MSBHA  | modular sensing bottomhole assembly |
| POOH   | pull out of hole                    |
| RIH    | run in hole                         |
| RTFOCT | real-time fiber-optic coiled tubing |
| SSD    | sliding sleeve door                 |
| SL     | slickline                           |
| WOB    | weight on bit                       |

## References

- Al Matar M., Mohapatra S., Al-Ateeqi H., et al 2020. Coiled Tubing-Assisted Real-Time Temperature Mapping for Discerning Completion Integrity Followed by Optimized Placement and Evaluation of a Porosity Fill Treatment for Water Shutoff in Dual-String Wells: Customized Solution. Presented at the SPE/ICoTA Well Intervention Conference and Exhibition, Houston, Texas, USA, 24-25 March. <https://doi.org/10.2118/199804-MS>.

- Al Sebea S.H., Abu-Eidah A.I., Patra, M., et al 2023. Enhanced Productivity Using Real-Time Coiled Tubing Assisted by Temperature Mapping and Integrated Dynamic Diversion Acid Squeeze Technique in a Highly Depleted Reservoir of West Kuwait. Presented at the SPE/ICoTA Well Intervention Conference and Exhibition, The Woodlands, Texas, USA, 21–22. <https://doi.org/10.2118/212924-MS>.
- Corrales, J., Ramirez, M., Tellez, J.R., et al 2024. The First Global Single-Run, Multiple Set-Unset Operation of an Electrically Activated Inflatable Packer Deployed with Hybrid Coiled Tubing: Case Study from Iraq where Real-Time Decisions on Selective Stimulation Treatments were made in a High-Permeability Contrast Reservoir. Presented at the ADIPEC, Abu Dhabi, UAE, 4-7 November. <https://doi.org/10.2118/221974-MS>.
- Daniels J. and Williams, G. 2001. High Density CCL Memory Tool to Confirm SSD Sleeve Position. Presented at the SPE/ICoTA Coiled Tubing Roundtable, Houston, Texas, USA, 7-8 March. <https://doi.org/10.2118/68362-MS>.
- Lerose, M., Pelusi L., Spagnolo, S., et al 2020. Real-Time Fiber-Optic Coiled Tubing Delivered Well Intervention Savings Through Faster Decision-Making Process. Presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 9-12 November. <https://doi.org/10.2118/202615-MS>.
- Vera, V., Torres, C., Delgado, E., et al 2019. Industry-First Hybrid Technology for Coiled Tubing Services Combining Fiber-Optic and Electric Line, Enabling Downhole Power and Real-Time Communication for the Next-Generation Decision-Making Process with Wider Downhole Insight. Presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 11-14 November. <https://doi.org/10.2118/197177-MS>.